Supplemental Material: Criticality in self-organized percolation

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S1. Growth rule

Fig. S1 illustrates the first steps of the growth, for site percolation on a small square lattice. The growth rule and Eq. (10) are described in the main text.

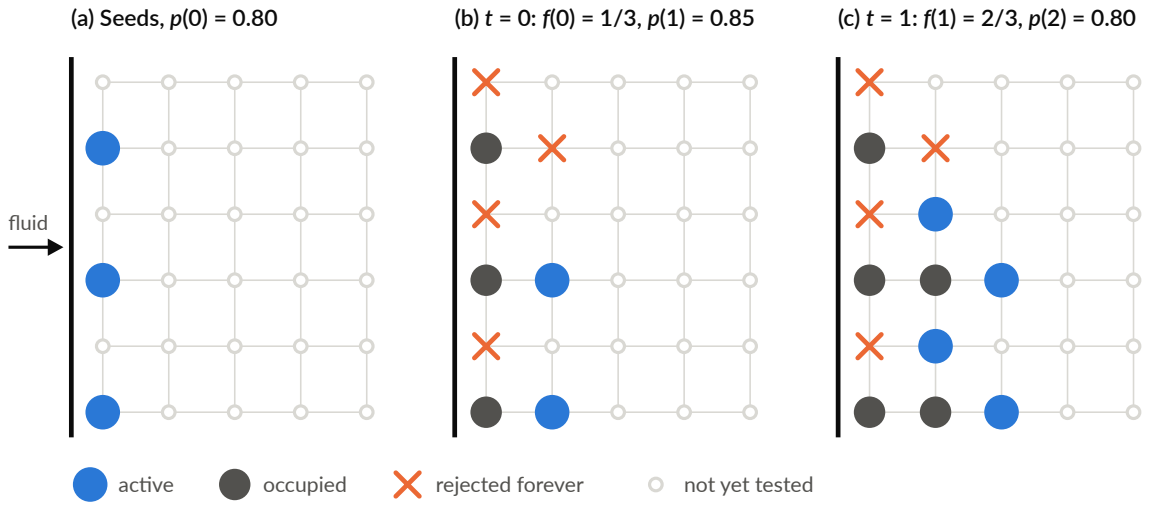


Figure S1: Site SOP on a small square lattice, with fluid entering through the left wall. (a) Three seeds are placed on the wall. (b) Each available neighbor of the seeds is tested once with probability $p(0)$: it is either filled and becomes active, or rejected forever. The fraction $f(0) = N(0)/L$ of active pores then sets $p(1)$ through Eq. (10). (c) The process repeats from the new active pores. Here $L = 6$, $c = 0.3$ and $f_T = 0.5$, and the states and values come from applying the rules directly.

S2. Independence of the inlet fraction

Fig. 2 shows that the stationary state does not depend on the initial value of p. Fig. S2 does the same for the seed density ρ_0, the fraction of the wall that starts occupied, with the feedback rule of Eq. (10). The curves nearly overlap from the first steps.

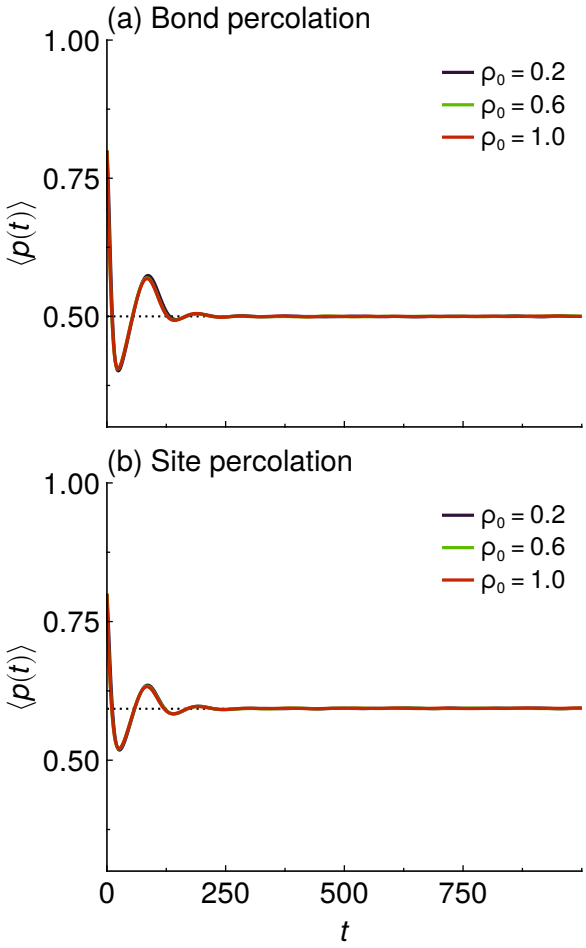


Figure S2: Independence of the inlet fraction. Mean $p(t)$ for (a) bond and (b) site percolation, with $L = 2^{14}$, $c = 0.05$, $f_T = 0.1$ and $p(0) = 0.8$, with seeds placed at random on the wall. Each curve is an average over 200 runs, none of which went extinct. Dotted lines mark p_c.

S3. Seed layout and boundary layer

The stationary state also does not depend on how the seeds are arranged on the wall, beyond a boundary layer. Fig. S3 compares random seeds with seeds grouped in one block and in 8 evenly spaced blocks, at the same ρ_0. Near the wall, blocks fill the medium more densely than random seeds, because a small front raises p through Eq. (10). Against the distance x in units of the gap g between blocks, the curves for both layouts and both sizes collapse: the difference from random seeds crosses zero near $x = 0.3g$, undershoots slightly, and vanishes near $x = g$. The layer is therefore about one gap thick, the distance over which the front has to spread sideways to cover the wall. Beyond it, the density, the plateau of p and the time to cross the lattice agree with those of random seeds. A clustered layout thus has a layer that grows with L, while random seeds, with gaps of a few sites, have none.

All layouts, random seeds included, also show a slow decay of the density from about 0.46 to 0.43 as x grows to about 10^3, so this second effect does not come from the gaps between blocks.

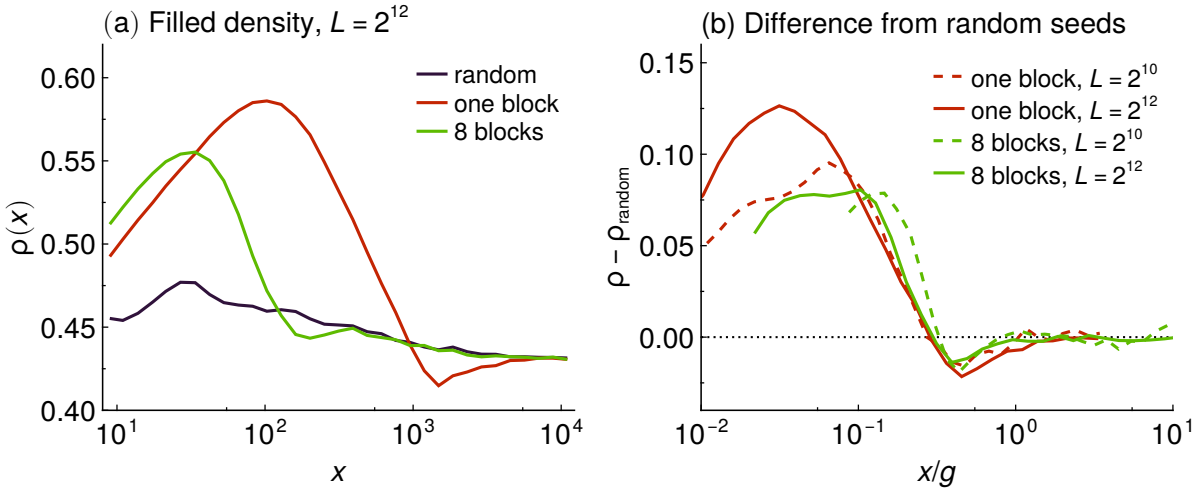


Figure S3: Boundary layer of the seed layout. Site percolation with $c = 0.05$, $f_T = 0.2$, $p(0) = 0.6$ and $\rho_0 = 0.2$, on a lattice of length $4L$, with periodic sides and 200 runs per case, all of which crossed the lattice. (a) Density of filled pores against the distance from the wall, for $L = 2^{12}$. (b) Difference from random seeds against x/g, where g is the empty stretch of wall between blocks, for $L = 2^{10}$ and $L = 2^{12}$. Dotted line: zero.

S4. The feedback as a discrete map

The stability of Eq. (10) is controlled by a single number. Neglecting exclusion effects, each active site produces on average p/p_c new active sites, so that, together with Eq. (10),

$$p(t+1) = p(t) + c[f_T - f(t)], \qquad f(t+1) = \frac{p(t+1)}{p_c} f(t). \tag{S1}$$

The fixed point is $(p, f) = (p_c, f_T)$. Linearizing around it, the deviations evolve with a matrix of determinant 1 and trace $2 - a$, where

$$a = \frac{c f_T}{p_c} = \frac{Q\tau F'(P^*)}{C p_c}. \tag{S2}$$

The second form uses Eq. (11), and it is the rise of p in one step with no pores being filled, in units of p_c. The eigenvalues are $e^{\pm i\theta}$, with

$$\cos\theta = 1 - \frac{a}{2}, \tag{S3}$$

so the deviations rotate around the fixed point for $a < 4$, and an eigenvalue reaches -1 at $a = 4$. For small a, the period is $T \approx 2\pi/\sqrt{a}$ steps.

The map differs from one-dimensional maps such as the logistic map in one respect: in the variables $(p, \ln f)$ its Jacobian has determinant 1 everywhere, not only at the fixed point, so it preserves area. The fixed point is then neutrally stable, and the map has no attractor. Any damping of the oscillations must come from what the map leaves out, such as exclusion effects and noise.

Table S1 compares Eq. (S3) with the lattice model, whose autocorrelation of p at lag k should be $\cos k\theta$ if the oscillation is a single rotation. The lag-1 values follow the map closely. The lag-2 values lie above the map at every a, so the lattice oscillation is slower or less coherent than a pure rotation. At $a = 2$, where $\theta = \pi/2$, the lattice shows a clear period-4 oscillation. The plateau p^* stays close to p_c only for small a, and it rises as the oscillations grow. All runs go extinct at the two values tested with $a \geq 3$, before the map reaches $a = 4$, most likely because the fluctuations are not small and p is bounded by $[0, 1]$. For the parameters of Fig. 2, $a = 0.01$.

a	c	Lag 1, map	Lag 1, lattice	Lag 2, map	Lag 2, lattice	p^*
0.02	0.05	0.990	0.995	0.960	0.980	0.5026
0.20	0.50	0.900	0.926	0.620	0.736	0.5022
0.50	1.25	0.750	0.789	0.125	0.314	0.5030
1.00	2.50	0.500	0.530	-0.500	-0.293	0.5031
1.50	3.75	0.250	0.260	-0.875	-0.695	0.5044
2.00	5.00	0.000	-0.021	-1.000	-0.875	0.5073
2.50	6.25	-0.250	-0.255	-0.875	-0.709	0.5541
3.00	7.50	-0.500	–	-0.500	–	–
3.50	8.75	-0.750	–	0.125	–	–

Table S1: Autocorrelation of $p(t)$ in the map, $\cos k\theta$ from Eq. (S3), and in the lattice model, against a. Bond percolation, $L = 4096$, $f_T = 0.2$, with 32 runs to $t = 6000$ starting from $p = p_c$ and a seed fraction f_T, measured over the second half of each run. Dashes mark values of a where all runs went extinct.

S5. Estimates for experiments

Table S2 gives the size of the parameters in practice, from Eq. (11) and Eq. (12), with $C = \beta V_c$ for a compressible liquid of volume V_c and $C = V_c/P_0$ for a gas at pressure P_0. With a gas in the system, the feedback is weak, since a in Eq. (S2) is below 10^{-3} at these rates. A pure liquid in a micromodel is the opposite case: a is well above the stable range of Table S1, because filling a single pore changes the pressure by more than the width of the thresholds, so the model does not apply. In practice the compliance of the tubing would soften such a system. The estimates of f_T are several orders of magnitude below the values simulated here. In a micromodel N_T is also below 1, so the invasion is a sequence of single bursts, not the synchronous front of the model. The viscous filling time is the weakest estimate: for water in a 1 mm pore it gives about 14 μs, whereas real filling events are slower, and a longer τ would raise f_T.

	Micromodel	Rock core
Pore size a	1 mm	30 μm
Pores in one layer, A	300	10^6
Pore volume of one layer, vA	3×10^{-7} m³	2.7×10^{-8} m³
Width of the thresholds, W	75 Pa	5000 Pa
Capillary number, $f_T \sim \mathrm{Ca}$	10^{-6} to 10^{-3}	10^{-7} to 10^{-5}
Pores filled per step, N_T	3×10^{-4} to 0.3	0.1 to 10
c, compressible liquid	9×10^5 (10 mL)	120 (100 mL)
c, gas	40 (10 mL at 1 atm)	0.5 (100 mL at 10 MPa)

Table S2: Order-of-magnitude estimates for slow drainage, for illustrative values of the parameters. The compliance uses the volume in brackets, with the compressibility of water for the liquid.